ITA0608 - MACHINE LEARNING LAB

PROGRAMS

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Experiment 4: Artificial Neural Network

Aim

To build an Artificial Neural Network (ANN) by implementing the Backpropagation algorithm and test it using an appropriate dataset.

Procedure

1. Import the required libraries.
2. Load an appropriate dataset.
3. Split the dataset into training and testing sets.
4. Normalize the input features.
5. Create an Artificial Neural Network model.
6. Add input, hidden, and output layers.
7. Train the network using the Backpropagation algorithm.
8. Test the trained model using the test dataset.
9. Evaluate the model accuracy.
10. Display the prediction results.

Algorithm

1. Start.
2. Load the dataset.
3. Preprocess the data (scaling and splitting).
4. Initialize the ANN with input, hidden, and output layers.
5. Assign random weights and biases.
6. Perform forward propagation to calculate outputs.
7. Compute the error between actual and predicted outputs.

8. Update weights and biases using Backpropagation.
9. Repeat until the stopping criterion is met.
10. Test the trained ANN using unseen data.
11. Calculate the accuracy.
12. Stop.

Code

```
import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

data = load_breast_cancer()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation='relu',
    solver='adam',
    max_iter=1000,
    random_state=42
)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

Output

```
Accuracy: 0.9824561403508771
Confusion Matrix:
[[42  1]
 [ 1 70]]
```

Result

The Artificial Neural Network was successfully built using the Backpropagation algorithm. The model was trained and tested on the Breast Cancer dataset and achieved high classification accuracy, demonstrating the effectiveness of Backpropagation for supervised learning.